

WIPRO CAPSTONE PROJECT

**PROJECT NAME: BIGBASKET WEB
APPLICATION AUTOMATION TESTING
USING SELENIUM WITH PYTEST
FRAMEWORK AND BDD USING BEHAVE
FRAMEWORK**

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MODULE NAME: Electronics

DOMAIN NAME: E-Commerce

Synopsis

PROJECT DESCRIPTION

SELENIUM WITH PYTEST

Introduction

Automation testing is used to improve software quality by reducing manual effort and increasing testing efficiency. Selenium WebDriver is an open-source automation testing tool used to automate web applications across different browsers. In this project, Selenium is integrated with the Pytest framework using Python programming language.

Pytest is a testing framework used for executing test cases, managing assertions, generating reports, and improving test organization. Selenium with Pytest helps in building scalable and reusable automation frameworks.

The framework automates various functionalities of the BigBasket web application such as:

- Login validation
- Electronics product search
- Product filtering
- Add to cart functionality
- Cart validation
- Product quantity update
- Checkout process

The framework follows the Page Object Model (POM) design pattern where locators, page methods, and test logic are separated for better maintainability and scalability.

The project also integrates:

- Allure Reports
- Screenshot capturing
- Logging
- Data-Driven Testing

to generate professional automation reports and improve debugging.

Advantages

- Reduces manual testing effort
 - Supports reusable automation scripts
 - Improves execution speed and accuracy
 - Generates professional reports
 - Supports regression testing
-

Conclusion

Selenium with Pytest provides a powerful automation testing solution for web applications. It improves framework maintainability, execution efficiency, and reporting capabilities through reusable components and structured framework design.

ARCHITECTURE

SELENIUM WITH PYTEST – FRAMEWORK ARCHITECTURE

Selenium_Project_Bigbasket

- |
- |— allure-report
 - | — index.html
- |— allure-results
 - | — screenshots
 - | — json files
- |— config
 - | — config.properties
- |— data
 - | — login_data.csv
- |— logs
 - | — automation.log
- |— pages
 - | — base_page.py
 - | — electronics_page.py
 - | — login_page.py
- |— tests
 - | — test_end-end.py
 - | — test_negative.py
 - | — test_positive.py
- |— utils
 - | — config_reader.py
 - | — csv_reader.py

```
| | — logger.py
| | — screenshot_util.py
| — conftest.py
— requirements.txt
```

TEST STRATEGY

Selenium Functional Testing

Objective

The objective of Selenium Functional Testing is to validate the core functionalities of the BigBasket web application using Selenium WebDriver with Python and Pytest framework.

What We Have Done

The following functional test scenarios were automated:

Positive Test Cases

- Login functionality validation
- Electronics product search
- Earbuds product filtering
- Add Earbuds product to cart
- Cart validation

Negative Test Cases

- Invalid mobile number validation
 - Invalid OTP validation
-

Framework Design

The Selenium framework was developed using:

- Python
 - Selenium WebDriver
 - Pytest Framework
 - Page Object Model (POM)
-

Additional Features

- Screenshot capturing
- Logging functionality
- CSV Data Handling
- Allure Reporting

TOOLS USED

Tool / Technology	Purpose
Python	Programming Language
Selenium WebDriver	Web Automation
Pytest	Test Execution Framework
Allure Reports	HTML Reporting
CSV File	Test Data Handling
Chrome/Edge Browser	Test Execution
PyCharm	IDE
GitHub	Version Control

TEST ENVIRONMENT

Operating System Used

- Windows 11
-

IDE Used

- PyCharm Community Edition
-

Tools Used

- Python
 - Selenium WebDriver
 - Pytest Framework
 - Allure Reports
 - CSV File
 - Chrome / Edge Browser
 - GitHub
-

requirements.txt Includes

- selenium
 - pytest
 - allure-pytest
 - webdriver-manager
-

FRAMEWORK ARCHITECTURE

The project follows a Hybrid Automation Framework architecture combining:

- Selenium Functional Framework
- Data-Driven Testing using CSV
- Page Object Model (POM)

Framework Components

- Page Classes
- Utility Files
- Test Files
- Config Files
- Reports
- Logs
- Screenshots

ALLURE REPORTS

Allure Reports are integrated for advanced HTML reporting.

Features Of Allure Reports

- Pass/Fail Status
- Step Execution Details
- Screenshot Attachments
- Error Logs
- Failure Analysis
- Execution Summary

Reports are automatically generated after test execution.

PAGE OBJECT MODEL (POM)

The framework follows the Page Object Model (POM) design pattern.

In POM:

- Locators are maintained separately
 - Reusable methods are maintained in page classes
 - Test logic is maintained in test files
-

Advantages Of POM

- Code Reusability
- Better Maintainability
- Reduced Code Duplication
- Improved Readability
- Scalable Framework Structure

TEST EXECUTION

End-To-End Testing Module

The End-To-End (E2E) Testing module validates the complete user workflow of the BigBasket web application from login to checkout process. The main objective of E2E testing is to ensure that all integrated modules of the application work correctly together without failures.

In this project, the End-To-End testing flow automates the complete shopping process including user login, product selection, cart operations, and checkout navigation. The automated E2E flow starts by logging into the application using valid credentials. After successful login, the user navigates to the Electronics category, selects Earbuds products, applies filters, adds products to the cart, and proceeds to checkout.

The End-To-End module ensures that:

- User navigation works correctly
- Product selection works properly
- Cart functionality behaves correctly
- Checkout flow is validated
- Product filters work correctly

This testing improves application reliability and verifies real-time user behavior across multiple modules.

TESTS CONDUCTED

A total of:

- Positive Test Cases
- Negative Test Cases
- End-To-End Test Scenarios

were executed during automation testing.

Positive Test Cases Conducted

- Login functionality validation
 - Electronics product search
 - Earbuds product filtering
 - Add Earbuds product to cart
 - Cart validation
-

Negative Test Cases Conducted

- Invalid mobile number validation
 - Invalid OTP validation
 - Invalid electronics product search
-

End-To-End Scenario Conducted

- User login
- Navigate to Electronics category
- Select Earbuds product
- Apply brand filter
- Add product to cart
- Cart validation
- Checkout process

The automated tests were executed successfully using Selenium WebDriver with Pytest framework. Allure Reports were generated for execution status, screenshots, and failure analysis.

TEST OUTPUTS

Outputs Screenshots:

TEST FLOW

Total Tests Executed : 7

- Positive Test Cases : 4
- Negative Test Cases : 2
- End-To-End Scenarios : 1

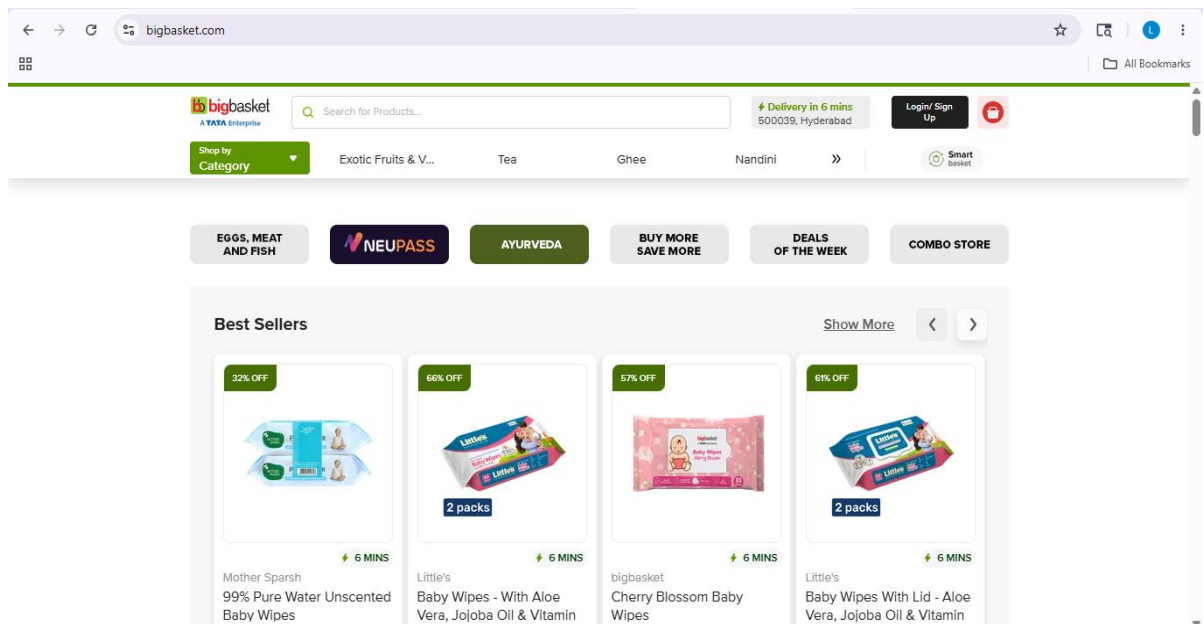
Selenium Functional Testing Flow

Step 1: Launch Browser

- Chrome / Edge browser is launched using Selenium WebDriver.

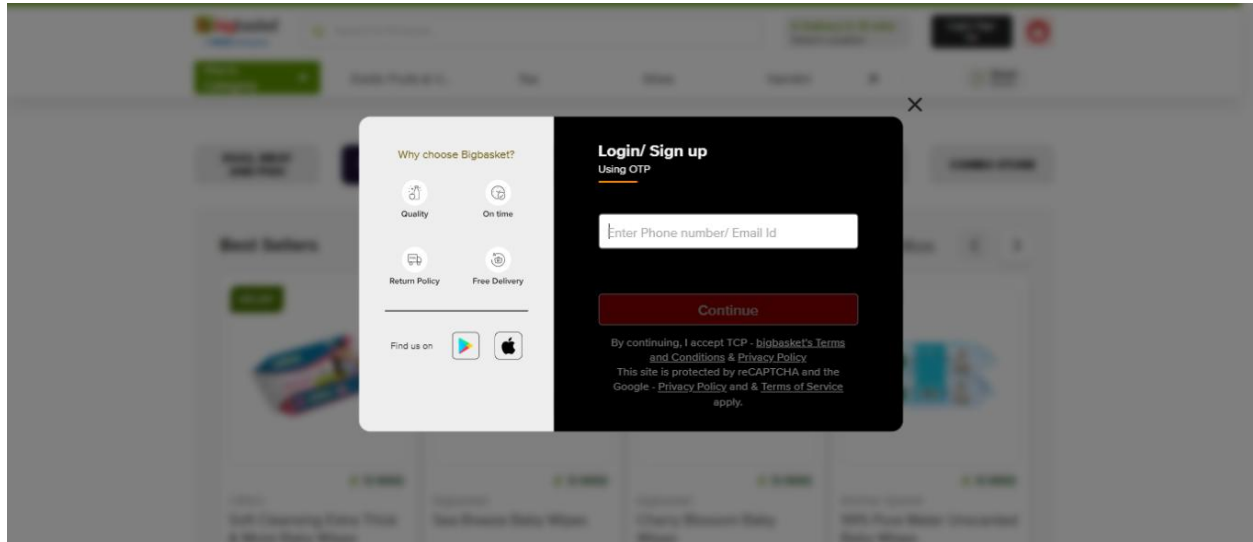
Step 2: Open BigBasket Application

- BigBasket website URL is opened.

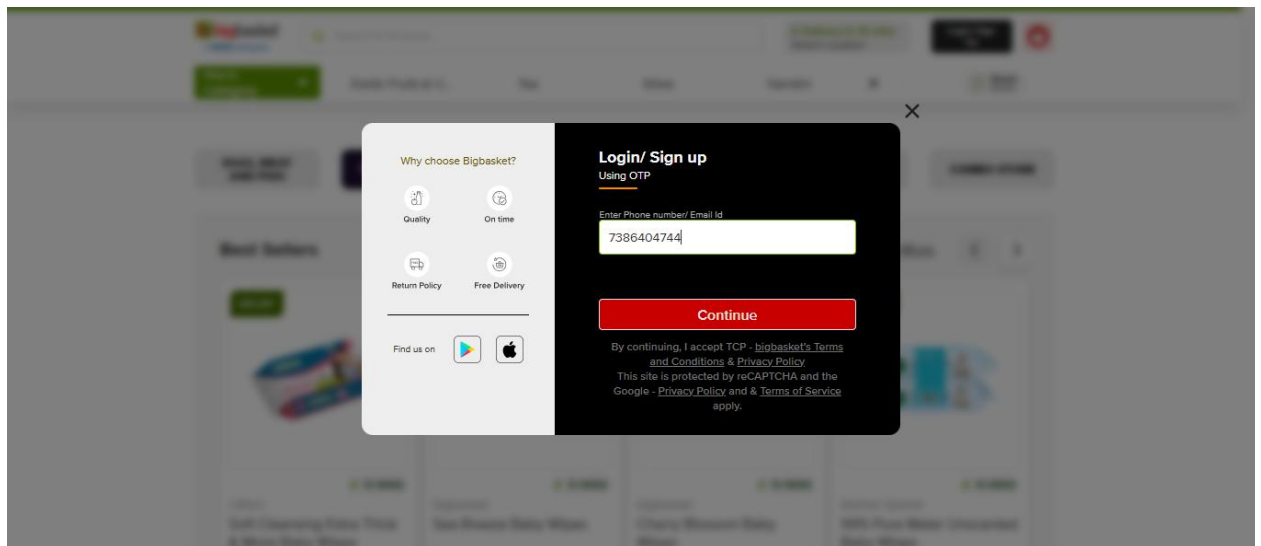


Step 3: User Login Validation

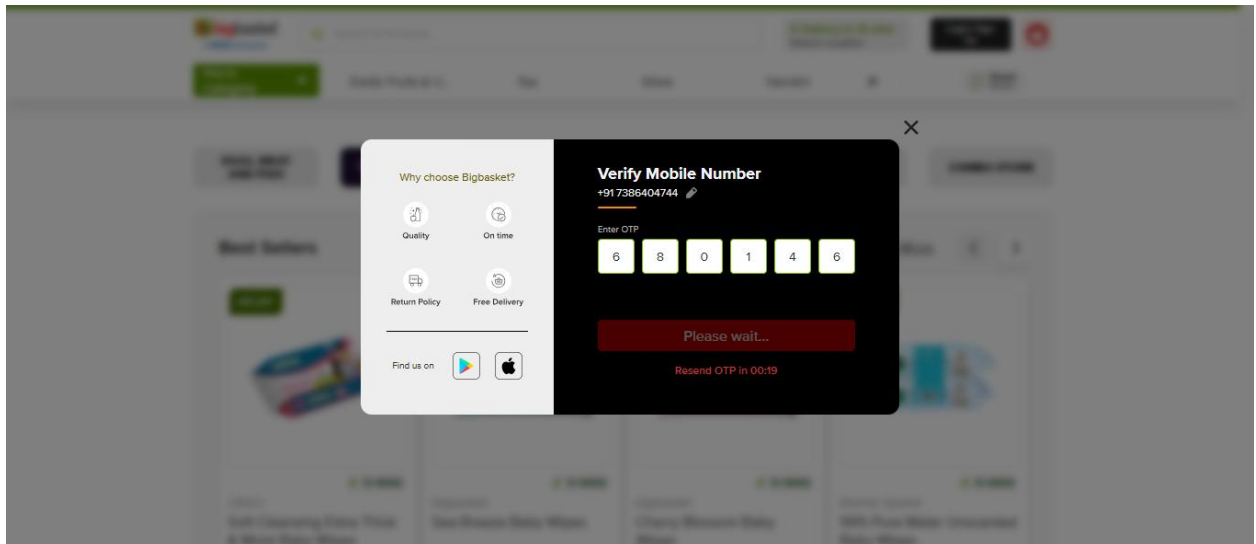
- User clicks login button



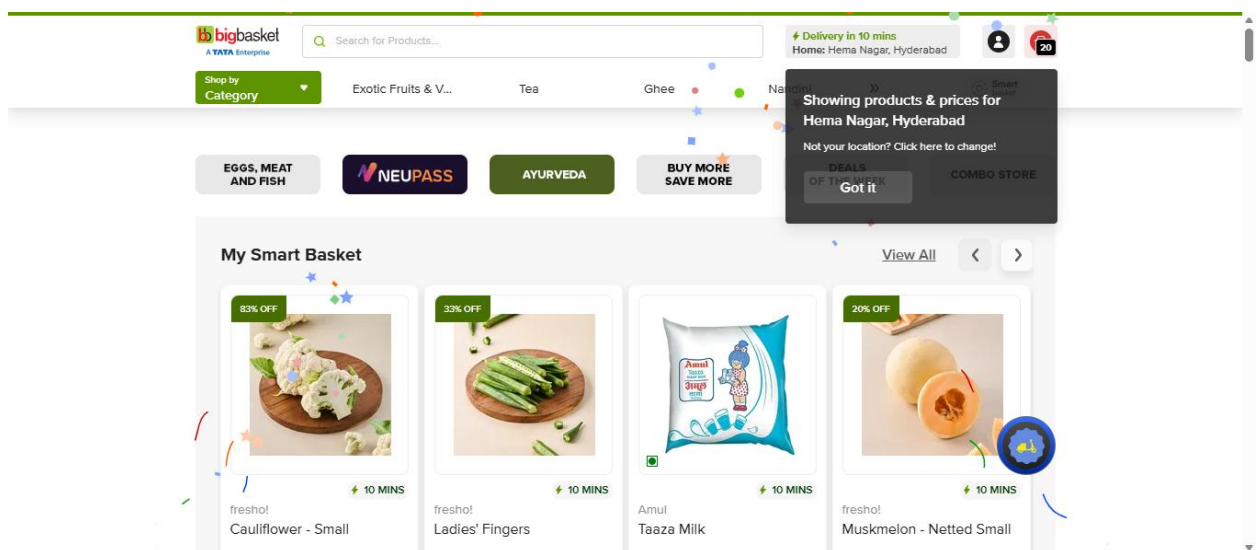
- Mobile number entered from csv file



- OTP verification

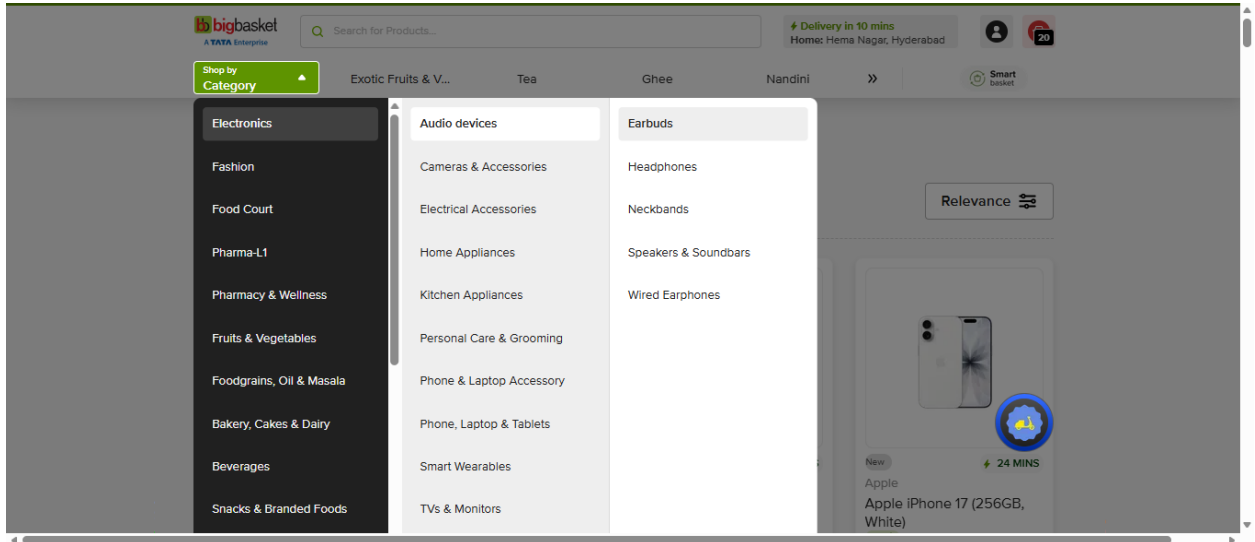


- Login success is validated

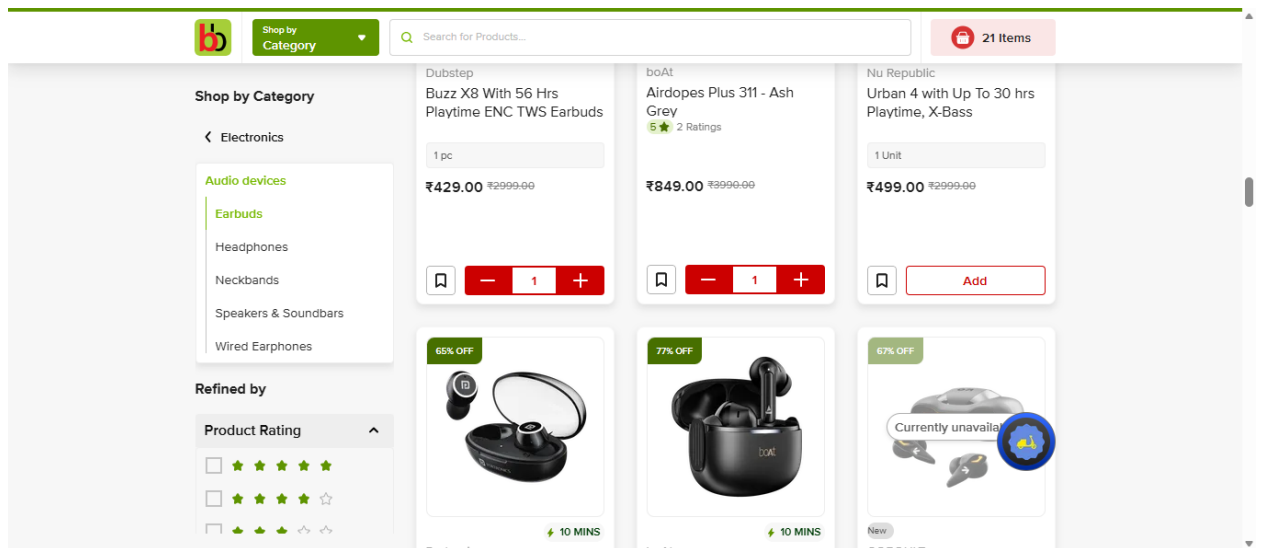


Step 4: Electronics Product Validation

- User navigates to Electronics category



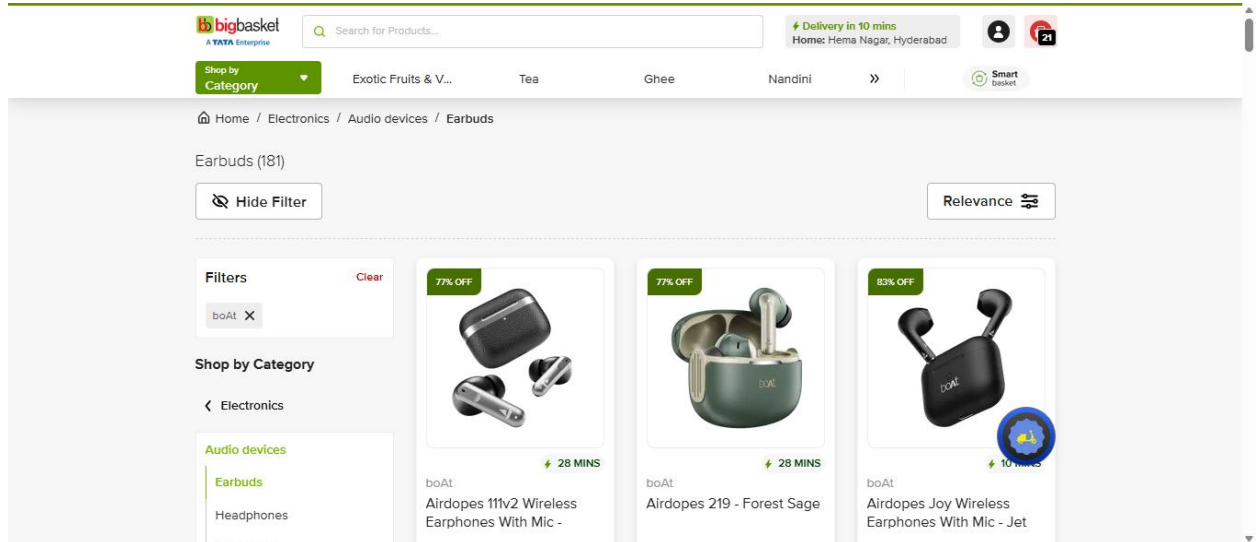
- Earbuds products are displayed



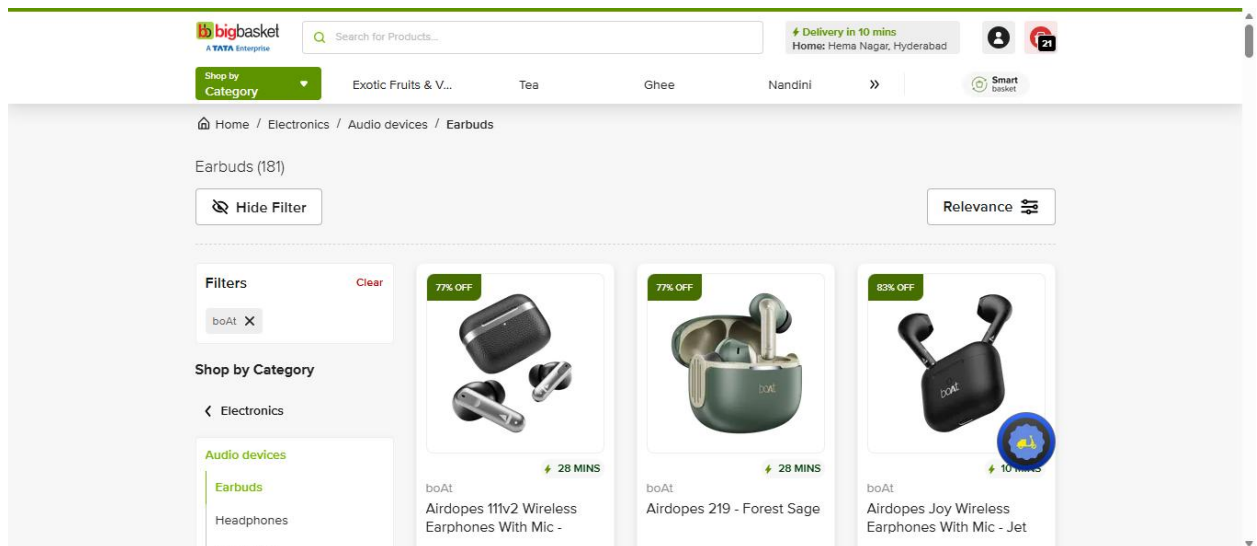
- Product search results are validated

Step 5: Product Filter Validation

- Brand filter is applied

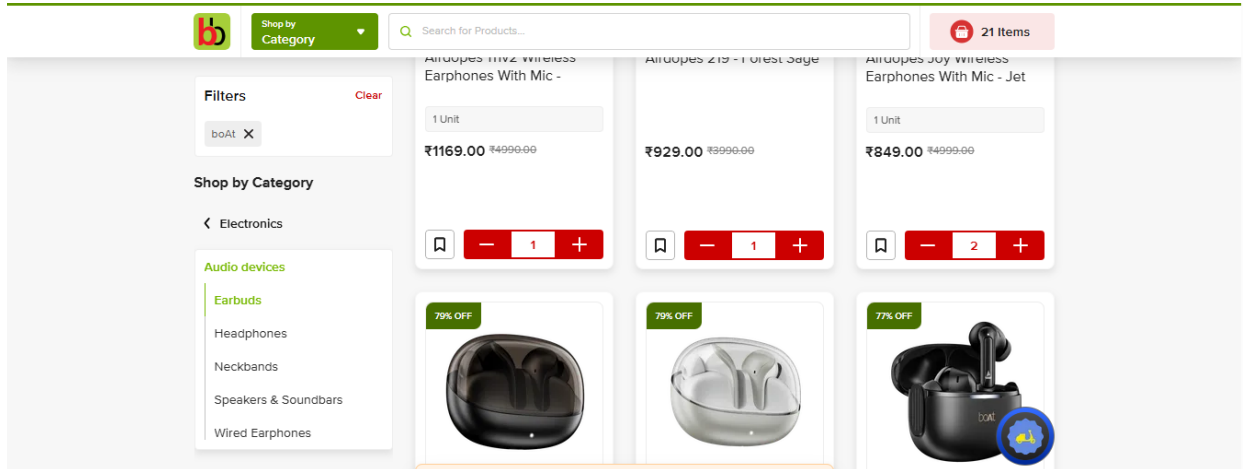


- Filtered products are displayed and validated



Step 6: Add Product To Cart

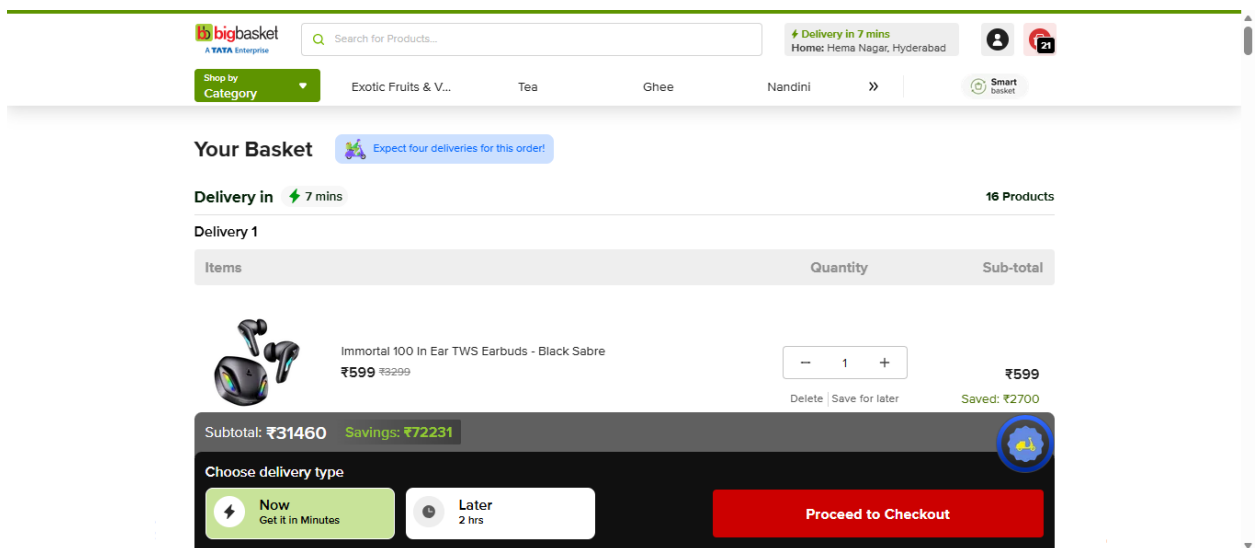
- Earbuds product is added to cart



- Cart item validation is performed

Step 7: Cart Validation

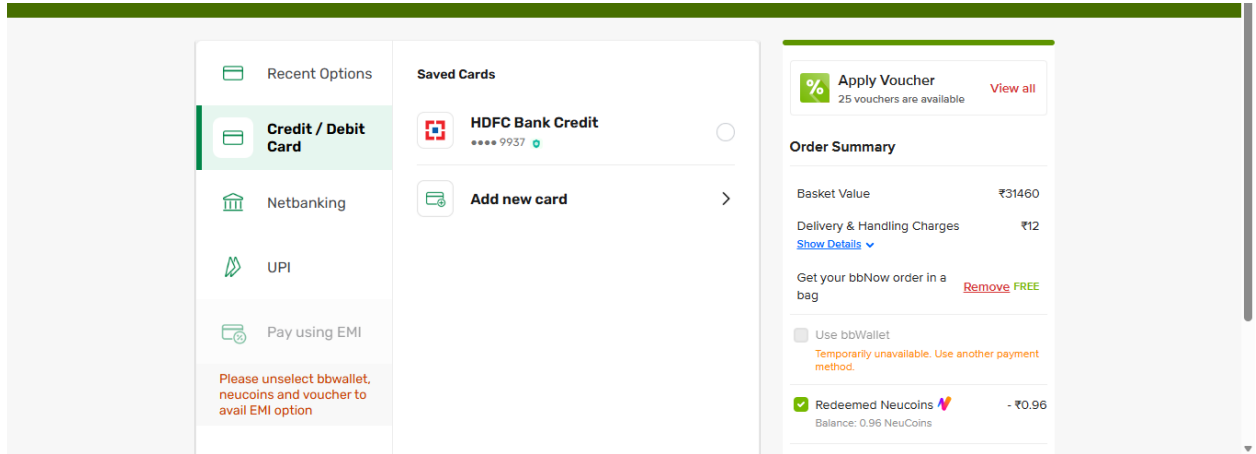
- User navigates to cart page



- Product availability in cart is validated

Step 8: Checkout Validation

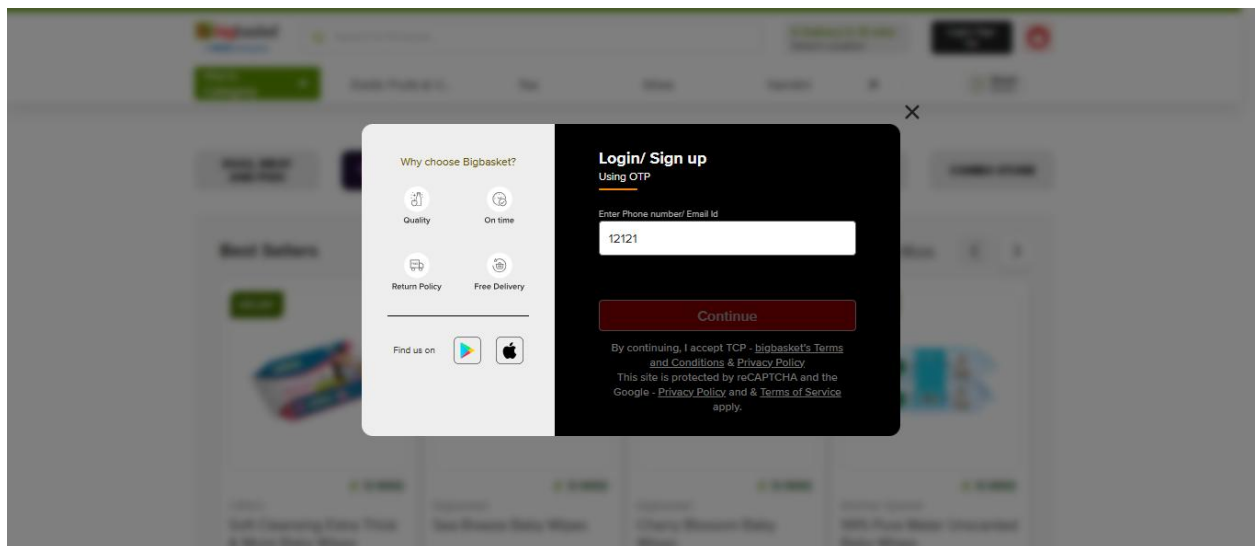
- User proceeds to checkout page



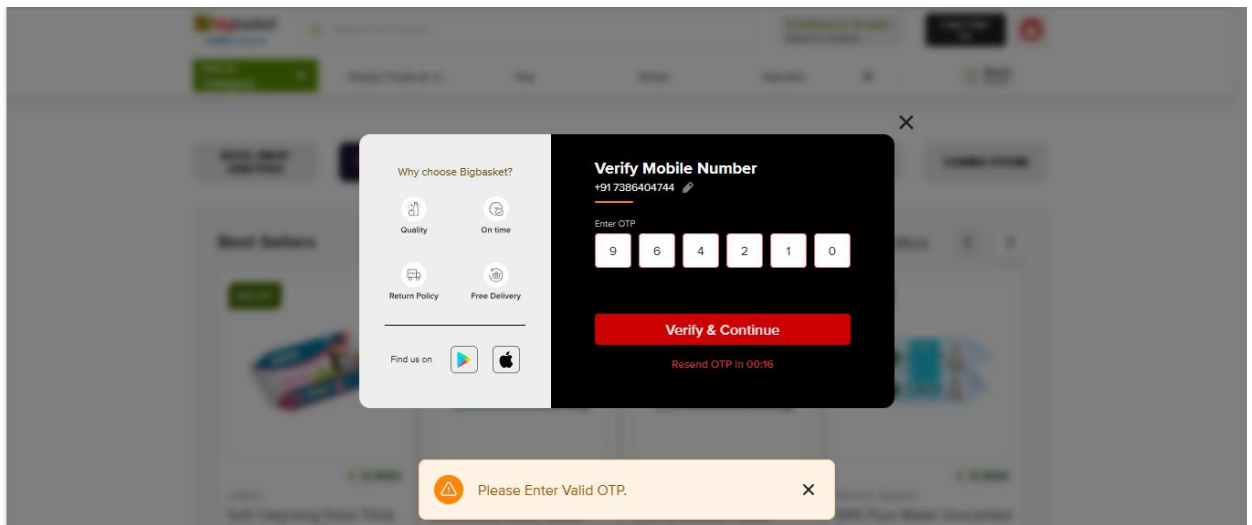
- Checkout navigation is validated

Step 9: Negative Validation

- Invalid mobile number validation



- Invalid OTP validation



LOGS

2026-05-23 17:33:15 - INFO - Launching browser: edge
2026-05-23 17:33:20 - INFO - BigBasket website opened
2026-05-23 17:33:46 - INFO - Login successful
2026-05-23 17:34:07 - INFO - Electronics category opened
2026-05-23 17:34:14 - INFO - Audio devices opened
2026-05-23 17:34:22 - INFO - Earbuds section opened
2026-05-23 17:34:49 - INFO - Boat brand selected
2026-05-23 17:34:57 - INFO - Product added to basket
2026-05-23 17:35:06 - INFO - Basket opened
2026-05-23 17:35:19 - INFO - Checkout page opened
2026-05-23 17:35:23 - INFO - Screenshot attached to Allure report
2026-05-23 17:35:23 - INFO - Automation logs attached to Allure report

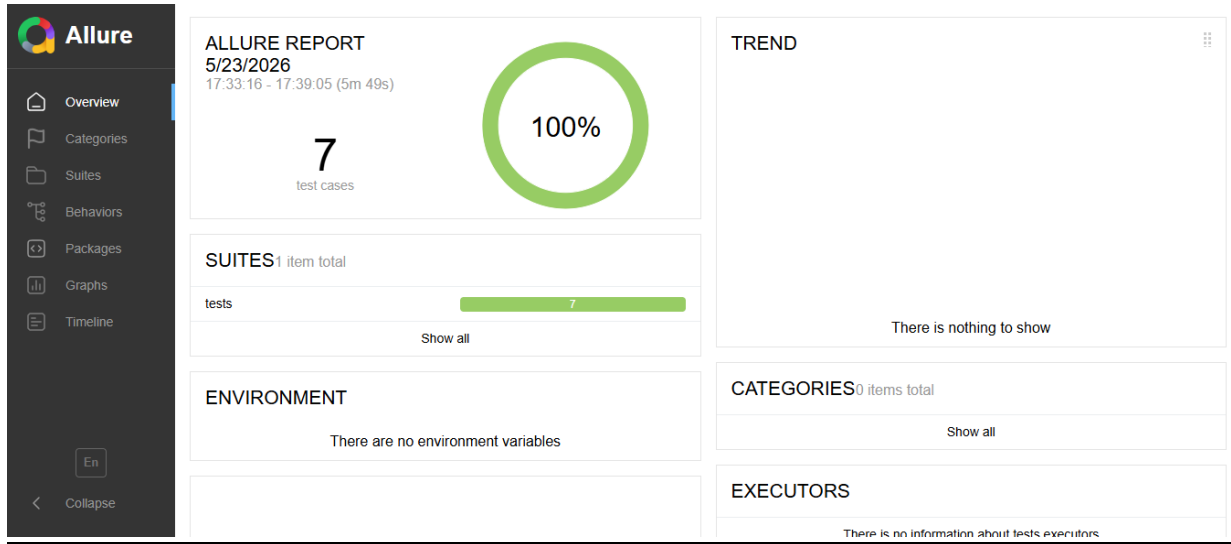
Negative Validation Logs:

2026-05-23 17:35:34 - INFO - Invalid mobile number entered
2026-05-23 17:35:34 - INFO - Verified Continue button is disabled
2026-05-23 17:35:34 - INFO - Invalid mobile test passed
2026-05-23 17:35:54 - INFO - Invalid OTP error message displayed
2026-05-23 17:35:54 - INFO - Invalid OTP test passed

Positive Validation Logs:

2026-05-23 17:36:19 - INFO - Login test passed
2026-05-23 17:37:02 - INFO - Open electronics category test passed
2026-05-23 17:37:58 - INFO - Apply filter test passed
2026-05-23 17:39:05 - INFO - Add product test passed

ALLURE-REPORT



The 'Suites' section provides a detailed view of the test suites. It includes a sidebar for navigation, a search bar, and a table of test suites.

order	name	duration	status	Status
0	0	0	7	0
1	0	0	0	0
2	0	0	0	0
3	0	0	0	0
4	0	0	0	0

Marks: [Icons]

▼ tests 7

- > test_end-end 1
- > test_negative 2
- > test_positive 4

Allure

Overview

Categories

Suites

Behaviors

Packages

Graphs

Timeline

En

Collapse

Suites

order ↑ name duration ↓ status ↓ Status: 0 0 7 0 0

Marks: [Icons]

tests 7

test_end-end 1

#1 BigBasket Electronics Flow (7386404744.) 2m 06s

test_negative 2

test_positive 4

Overview History Retries

Severity: critical

Duration: 2m 06s

Parameters

mobile: ('7386404744.')

Execution

Set up

Test body

test_bigbasket_electronics_flow[mobile0] Screenshot 51 KB

Automation Logs 1.1 KB

2026-05-23 17:33:15,136 - INFO - Launching browser: edge

2026-05-23 17:33:16,419 - INFO - Browser launched successfully

2026-05-23 17:33:20,330 - INFO - BigBasket website opened

2026-05-23 17:33:24,116 - INFO - Login button clicked

2026-05-23 17:33:26,397 - INFO - Mobile number entered

2026-05-23 17:33:26,726 - INFO - Continue button clicked

2026-05-23 17:33:46,069 - INFO - Login successful

2026-05-23 17:33:51,385 - INFO - Got It popup handled

2026-05-23 17:33:57,525 - INFO - Shop By Category opened

2026-05-23 17:34:07,702 - INFO - Electronics category opened

2026-05-23 17:34:14,855 - INFO - Audio devices opened

2026-05-23 17:34:22,028 - INFO - Earbuds section opened

2026-05-23 17:34:36,226 - INFO - First earbud quantity increased

2026-05-23 17:34:43,312 - INFO - Brand filter opened

2026-05-23 17:34:49,923 - INFO - Boat brand selected

2026-05-23 17:34:57,045 - INFO - Product added to basket

Allure

Overview

Categories

Suites

Behaviors

Packages

Graphs

Timeline

En

Collapse

Suites

order ↑ name duration ↓ status ↓ Status: 0 0 7 0 0

Marks: [Icons]

tests 7

test_end-end 1

test_negative 2

#1 Invalid Mobile Number 5s 777ms

#2 Invalid OTP Login 16s 173ms

test_positive 4

No item selected

Allure

Overview

Categories

Suites

Behaviors

Packages

Graphs

Timeline

En

Collapse

Suites

order ↑ name duration ↓ status ↓ Status: 0 0 7 0 0

Marks: [Icons]

tests 7

test_end-end 1

test_negative 2

#1 Invalid Mobile Number 5s 777ms

#2 Invalid OTP Login 16s 173ms

test_positive 4

tests.test_negative#test_invalid_mobile

Passed Invalid Mobile Number

Overview History Retries

Severity: normal

Duration: 5s 777ms

Execution

Set up

Test body

invalid_mobile 97.9 KB

test_invalid_mobile_Screenshot 97.9 KB

Overview

Categories

Suites

Behaviors

Packages

Graphs

Timeline

En

Collapse

Suites

order ↑ name duration ↓ status ↓

Status: 0 0 7 0 0

Marks: [Icons]

tests 7

test_end-end 1

test_negative 2

#1 Invalid Mobile Number 5s 777ms

#2 Invalid OTP Login 16s 173ms

test_positive 4

invalid_mobile 97.9 KIB

test_invalid_mobile_Screenshot 97.9 KIB

Automation Logs 1.9 KIB

2026-05-23 17:33:15,136 - INFO - Launching browser: edge
2026-05-23 17:33:16,419 - INFO - Browser launched successfully
2026-05-23 17:33:20,330 - INFO - BigBasket website opened
2026-05-23 17:33:24,116 - INFO - Login button clicked
2026-05-23 17:33:26,397 - INFO - Mobile number entered
2026-05-23 17:33:26,726 - INFO - Continue button clicked
2026-05-23 17:33:46,869 - INFO - Login successful
2026-05-23 17:33:51,385 - INFO - Got It popup handled
2026-05-23 17:33:57,525 - INFO - Shop By Category opened
2026-05-23 17:34:07,782 - INFO - Electronics category opened
2026-05-23 17:34:14,855 - INFO - Audio devices opened
2026-05-23 17:34:22,028 - INFO - Earbuds section opened
2026-05-23 17:34:36,226 - INFO - First earbud quantity increased
2026-05-23 17:34:43,312 - INFO - Brand filter opened

Overview

Categories

Suites

Behaviors

Packages

Graphs

Timeline

En

Collapse

Suites

order ↑ name duration ↓ status ↓

Status: 0 0 7 0 0

Marks: [Icons]

tests 7

test_end-end 1

test_negative 2

test_positive 4

#4 test_add_product 1m 03s

#3 test_apply_filter 51s 532ms

#1 test_login 21s 268ms

#2 test_open_electronics 38s 991ms

No item selected

Overview

Categories

Suites

Behaviors

Packages

Graphs

Timeline

En

Collapse

Suites

order ↑ name duration ↓ status ↓

Status: 0 0 7 0 0

Marks: [Icons]

tests 7

test_end-end 1

test_negative 2

test_positive 4

#4 test_add_product 1m 03s

#3 test_apply_filter 51s 532ms

#1 test_login 21s 268ms

#2 test_open_electronics 38s 991ms

tests.test_positive#test_login

Passed test_login

Overview History Retries

Severity: normal

Duration: 21s 268ms

Execution

Set up

Test body

test_login_Screenshot 93.1 KIB

Automation Logs 3.6 KIB

Overview

Categories

Suites

Behaviors

Packages

Graphs

Timeline

En

Collapse

Suites

order ↑ name duration ↓ status ↓ Status: 0 0 7 0 0

Marks: [Icons]

tests 7

test_end-end 1

test_negative 2

test_positive 4

#4 test_add_product 1m 03s

#3 test_apply_filter 51s 532ms

#1 test_login 21s 268ms

#2 test_open_electronics 38s 991ms

tests.test_positive#test_open_electronics

Passed test_open_electronics

Overview History Retries

Severity: normal

Duration: 38s 991ms

Execution

Set up

Test body

test_open_electronics_Screenshot 68.7 KIB

Automation Logs 4.5 KIB

2026-05-23 17:33:15.136 - INFO - Launching browser: edge

Overview

Categories

Suites

Behaviors

Packages

Graphs

Timeline

En

Collapse

Suites

order ↑ name duration ↓ status ↓ Status: 0 0 7 0 0

Marks: [Icons]

tests 7

test_end-end 1

test_negative 2

test_positive 4

#4 test_add_product 1m 03s

#3 test_apply_filter 51s 532ms

#1 test_login 21s 268ms

#2 test_open_electronics 38s 991ms

tests.test_positive#test_apply_filter

Passed test_apply_filter

Overview History Retries

Severity: normal

Duration: 51s 532ms

Execution

Set up

Test body

test_apply_filter_Screenshot 110.9 KIB

Automation Logs 5.5 KIB

2026-05-23 17:33:15.136 - INFO - Launching browser: edge

Overview

Categories

Suites

Behaviors

Packages

Graphs

Timeline

En

Collapse

Suites

order name duration status

Status: 0 0 7 0 0

Marks:

tests

> test_end-end

> test_negative

> test_positive

✓ #4 test_add_product

✓ #3 test_apply_filter

✓ #1 test_login

✓ #2 test_open_electronics

tests.test_positive#test_add_product

Passed test_add_product

Overview History Retries

Severity: normal

Duration: 1m 03s

Execution

> Set up

> Test body

test_add_product_Screenshot

Automation Logs

Overview

Categories

Suites

Behaviors

Packages

Graphs

Timeline

En

Collapse

STATUS

100%

Failed

Broken

Passed

Skipped

Unknown

SEVERITY

DURATION

DURATION TREND

TEST REPORT

Selenium With Pytest

The Selenium Functional Testing framework was executed successfully using the Pytest framework integrated with Selenium WebDriver. The framework automated major functionalities of the BigBasket web application including login validation, Electronics product navigation, Earbuds product filtering, cart operations, and checkout validation.

Total Test Cases Executed

- 7 Test Cases
-

Test Execution Result

- Passed Test Cases : 7
 - Failed Test Cases : 0
 - Skipped Test Cases : 0
-

Accuracy Achieved

- Test Execution Accuracy : 100%
-

Functional Modules Tested

- Login Functionality
- Electronics Category Navigation
- Earbuds Product Validation
- Product Filter Validation
- Add Product To Cart
- Cart Validation

- Checkout Validation
- Invalid Input Validation

Reporting And Validation

During execution:

- Screenshots were captured automatically
- Logs were generated successfully
- Allure HTML Reports were generated
- Assertions validated application behavior

The Selenium with Pytest framework execution completed successfully without failures, and all validations passed as expected.

CONCLUSION

The BigBasket Automation Testing Framework was successfully developed and implemented using Selenium WebDriver with Pytest framework. The framework automated major functionalities of the BigBasket web application including login validation, Electronics category navigation, Earbuds product filtering, cart operations, and checkout validation.

The project successfully demonstrated:

- Functional Testing
- Negative Testing
- End-To-End Testing
- Assertion Validation
- Reporting And Logging

The framework was designed using the Page Object Model (POM) architecture, which improved code reusability, readability, scalability, and maintainability. Integration of Allure Reports, screenshots, and logging mechanisms provided professional reporting and easier debugging capabilities.

Using Pytest improved test execution management, assertion handling, and automation workflow. The End-To-End testing module validated the complete user workflow from login to checkout process successfully.

All automated test cases executed successfully with high accuracy, validating the stability and functionality of the application. The framework reduced manual testing effort, improved execution efficiency, and provided a scalable automation solution suitable for real-time industry projects.

Overall, the project successfully achieved its objective of developing a reliable, maintainable, and professional Selenium automation testing framework for web application testing.

Synopsis

PROJECT DESCRIPTION

BDD WITH BEHAVE

Introduction

Behavior Driven Development (BDD) is a software testing methodology where test scenarios are written in simple English language using Gherkin syntax. Behave is the Python BDD framework used in this project along with Selenium WebDriver.

BDD improves collaboration between developers, testers, and business stakeholders by making test cases easy to understand and readable.

Feature files are written using:

- Given
- When
- Then

keywords.

Example

Scenario: Add Earbuds Product To Cart

Scenario: Add Earbuds Product To Cart

Given user is logged into BigBasket

When the user navigates to Electronics category

And the user selects Earbuds product

Then the Earbuds product should be added successfully

The framework automates:

- Positive test scenarios
- Negative test scenarios
- End-To-End testing scenarios

using reusable step definitions and Page Object Model (POM) design pattern.

The project also includes:

- Allure Reporting
- Screenshot attachments
- Logging functionality

to improve reporting and debugging.

Conclusion

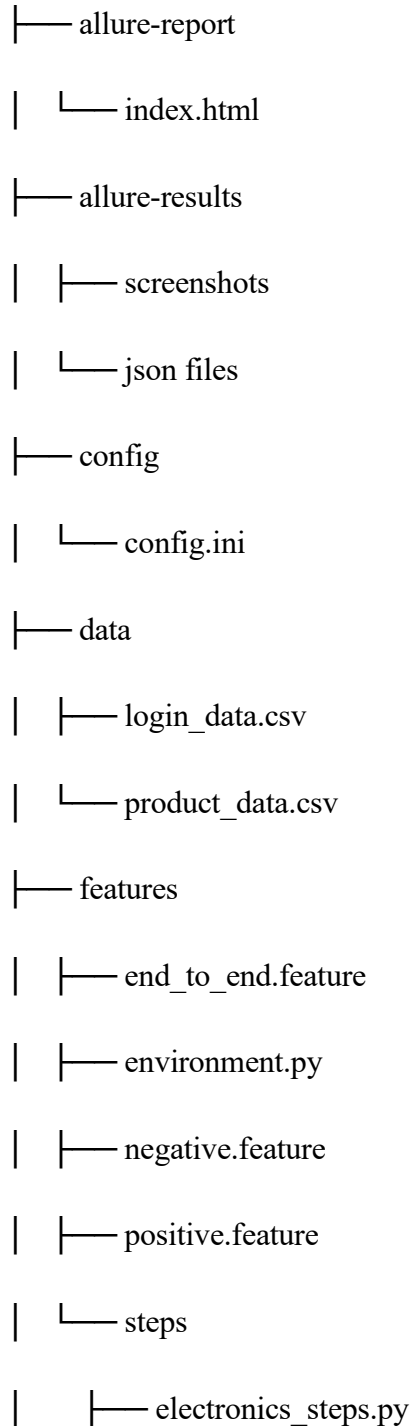
BDD with Behave provides a readable and maintainable automation testing framework.

Integration with Selenium improves automation efficiency while Allure Reports, screenshots, and logging provide professional reporting and debugging support.

ARCHITECTURE

BDD WITH BEHAVE – FRAMEWORK ARCHITECTURE

BDD_Project_Bigbasket



```
|   └─ login_steps.py
|
└─ locators
|   └─ electronics_locators.py
|   └─ login_locators.py
|
└─ logs
|   └─ automation.log
|
└─ pages
|   └─ base_page.py
|   └─ electronics_page.py
|   └─ login_page.py |
|
└─ utils
|   └─ config_reader.py
|   └─ csv_reader.py
|   └─ logger.py
|   └─ screenshot_util.py
|   └─ waits.py
|
└─ behave.ini
|
└─ generate_html_report.py
|
└─ requirements.txt
|
└─ runtest.py
```

TEST STRATEGY

BDD Functional Testing

Objective

The objective of BDD Functional Testing is to validate application behavior using simple English-readable test scenarios with Behave framework.

What We Have Done

BDD scenarios were created for:

- Login validation
 - Electronics category validation
 - Earbuds product filtering
 - Add product to cart
 - Cart functionality
 - Checkout process
 - End-To-End shopping flow
-

BDD Implementation

The framework was developed using:

- Behave BDD Framework
- Gherkin Feature Files

- Step Definitions
 - Selenium WebDriver
 - Page Object Model
-

End-To-End Flow Automated

- User login
 - Navigate to Electronics category
 - Open Audio Devices section
 - Select Earbuds products
 - Apply Boat brand filter
 - Add product to cart
 - Navigate to basket
 - Proceed to checkout
-

Additional Features

- Allure Reports
- Screenshot attachments
- Logging support
- Reusable step definitions

TOOLS USED

Tool / Technology	Purpose
Python	Programming Language
Selenium WebDriver	Web Automation
Behave	BDD Framework
Gherkin	Scenario Writing
Allure Reports	HTML Reporting
CSV File	Test Data Handling
WebDriver Manager	Browser Driver Management
Chrome/Edge Browser	Test Execution
PyCharm	IDE
GitHub	Version Control

TEST ENVIRONMENT

Operating System Used

- Windows 11
-

IDE Used

- PyCharm Community Edition
-

Framework Used

- Behave BDD Framework
-

Tools Used

- Python
 - Selenium WebDriver
 - Behave
 - Gherkin
 - Allure Reports
 - CSV File
 - Chrome / Edge Browser
 - GitHub
-

requirements.txt Includes

- selenium
- behave
- allure-behave
- webdriver-manager

FRAMEWORK ARCHITECTURE

The project follows a Hybrid Automation Framework architecture combining:

- BDD Functional Framework
- Data-Driven Testing using CSV
- Page Object Model (POM)
- End-To-End Automation Testing

Framework Components

- Feature Files
 - Step Definition Files
 - Locator Files
 - Page Classes
 - Utility Files
 - Test Data Files
 - Reports
 - Logs
 - Screenshots
-

ALLURE REPORTS

Allure Reports are integrated for advanced HTML reporting.

Features Of Allure Reports

- Pass/Fail Status
- Step Execution Details
- Screenshot Attachments
- Error Logs
- Failure Analysis
- Execution Summary

Reports are automatically generated after test execution.

PAGE OBJECT MODEL (POM)

The framework follows the Page Object Model (POM) design pattern.

In POM:

- Locators are maintained separately
 - Reusable methods are maintained in page classes
 - Test logic is maintained in step definition files
-

Advantages Of POM

- Code Reusability
- Better Maintainability
- Reduced Code Duplication
- Improved Readability
- Scalable Framework Structure

TEST EXECUTION

End-To-End Testing Module

The End-To-End (E2E) Testing module validates the complete user workflow of the BigBasket web application from login to checkout process. The main objective of E2E testing is to ensure that all integrated modules of the application work correctly together without failures.

In this project, the End-To-End testing flow automates the complete shopping process including user login, Electronics category navigation, product filtering, cart operations, and checkout validation.

The automated E2E flow starts by logging into the application using valid credentials. After successful login, the user navigates to the Electronics category, opens Audio Devices, selects Earbuds products, applies brand filters, adds products to the basket, and proceeds to checkout.

The End-To-End module ensures that:

- User navigation works correctly
- Product selection works properly
- Cart functionality behaves correctly
- Checkout flow is validated
- Product filters work correctly

This testing improves application reliability and verifies real-time user behavior across multiple modules.

TESTS CONDUCTED

A total of:

- Positive Test Cases
- Negative Test Cases
- End-To-End Test Scenarios

were executed during automation testing.

Positive Test Cases Conducted

- Login functionality validation
 - Electronics category validation
 - Earbuds product filtering
 - Add product to basket
 - Cart validation
-

Negative Test Cases Conducted

- Invalid mobile number validation
 - Invalid OTP validation
-

End-To-End Scenario Conducted

- User login
- Navigate to Electronics category
- Open Audio Devices section
- Select Earbuds products
- Apply Boat brand filter
- Add product to basket
- Basket validation
- Checkout process

The automated tests were executed successfully using Selenium WebDriver with Behave BDD framework. Allure Reports were generated for execution status, screenshots, and failure analysis.

TEST OUTPUTS

Outputs Screenshots

TEST FLOW

Total Tests Executed : 7

- Positive Test Cases : 4
 - Negative Test Cases : 2
 - End-To-End Scenarios : 1
-

BDD Functional Testing Flow

Step 1: Create Feature Files

- Test scenarios are written in Gherkin language using:
 - Given
 - When
 - Then
-

Step 2: Create Step Definitions

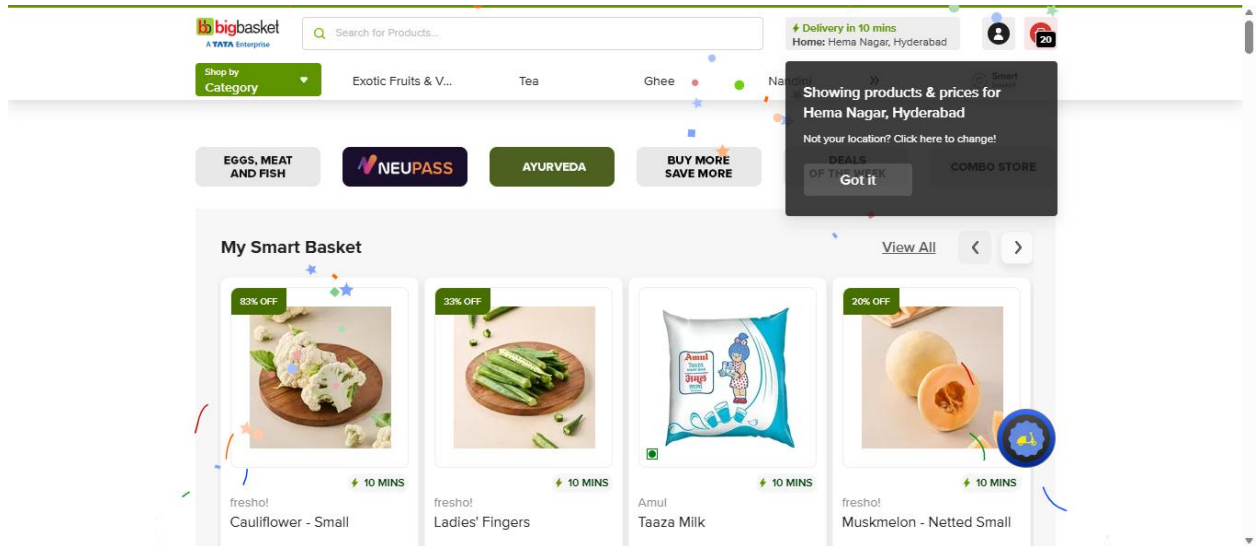
- Step implementation is written using Behave framework.
-

Step 3: Launch Browser

- Selenium WebDriver launches Chrome / Edge browser.
-

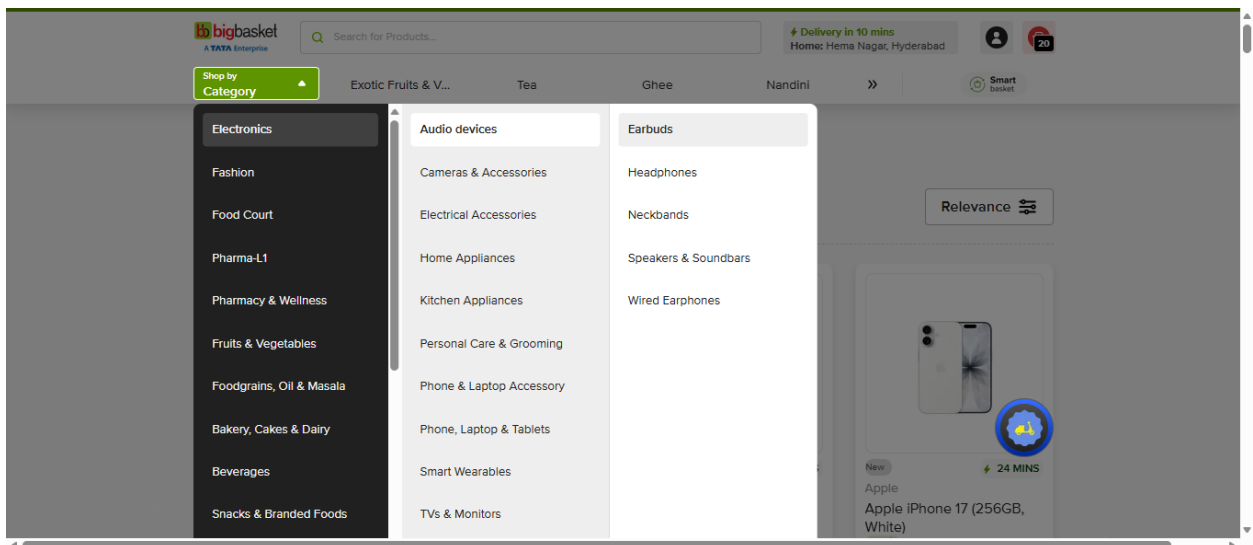
Step 4: Execute Login Scenario

- User login flow is automated and validated.

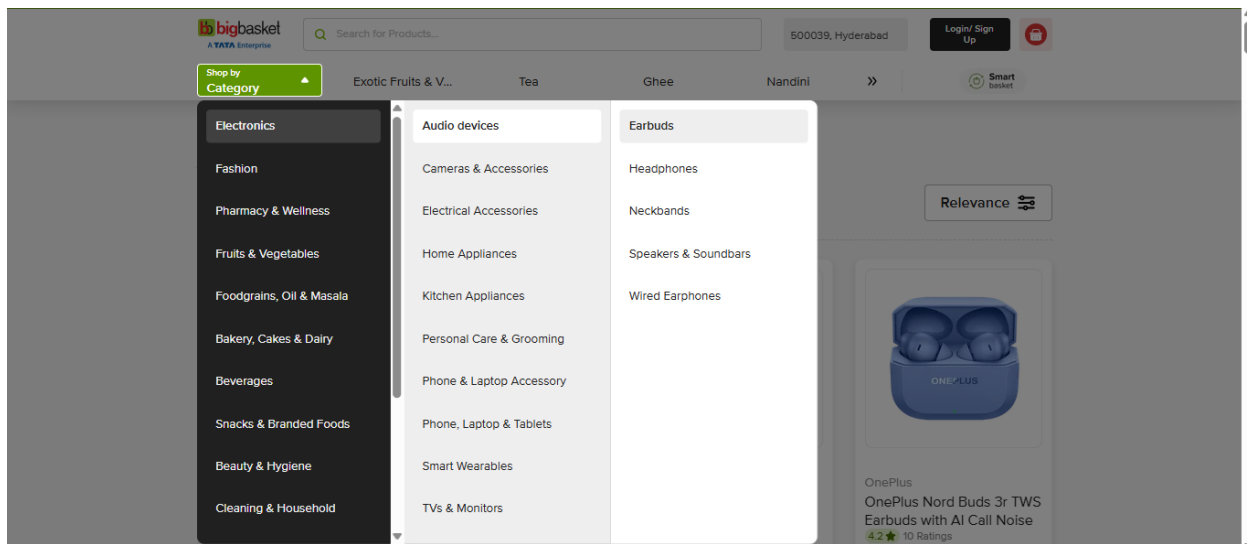


Step 5: Execute Electronics Scenarios

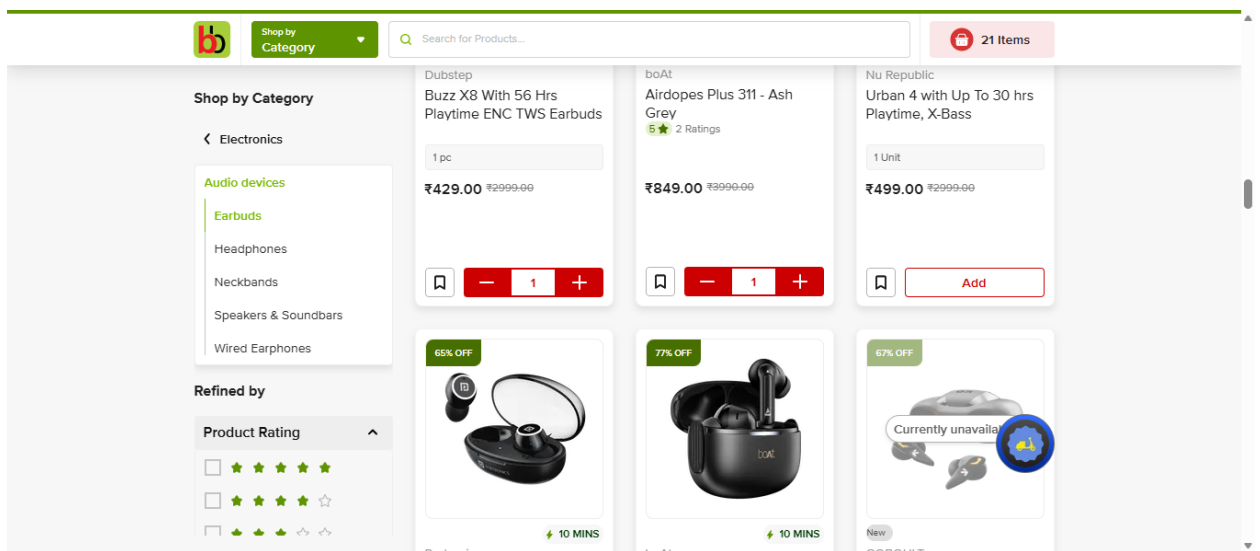
- Electronics category navigation



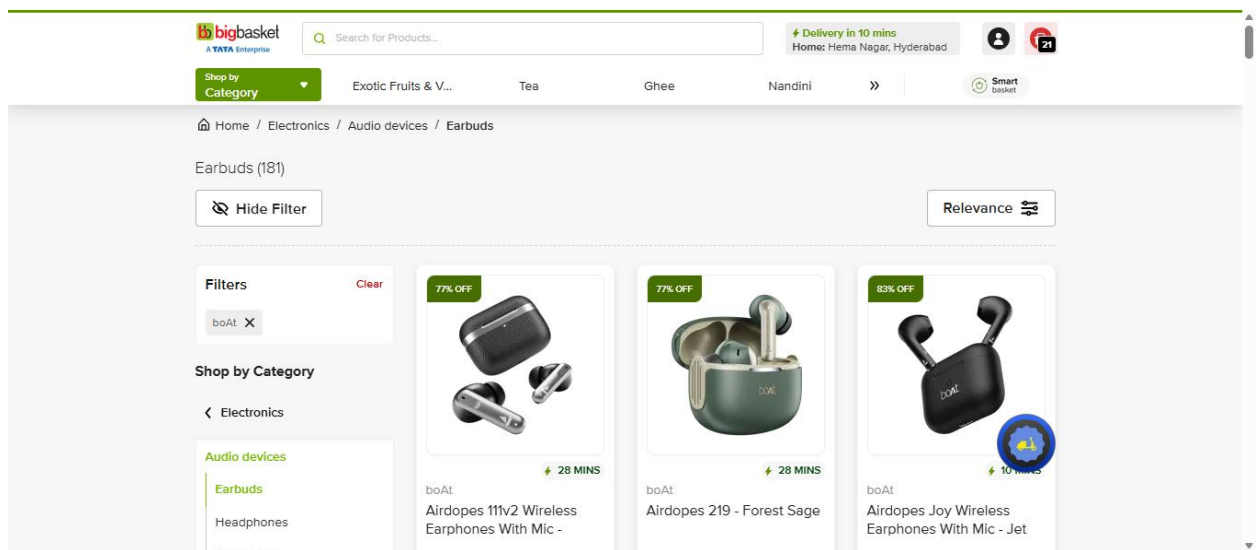
- Audio Devices navigation



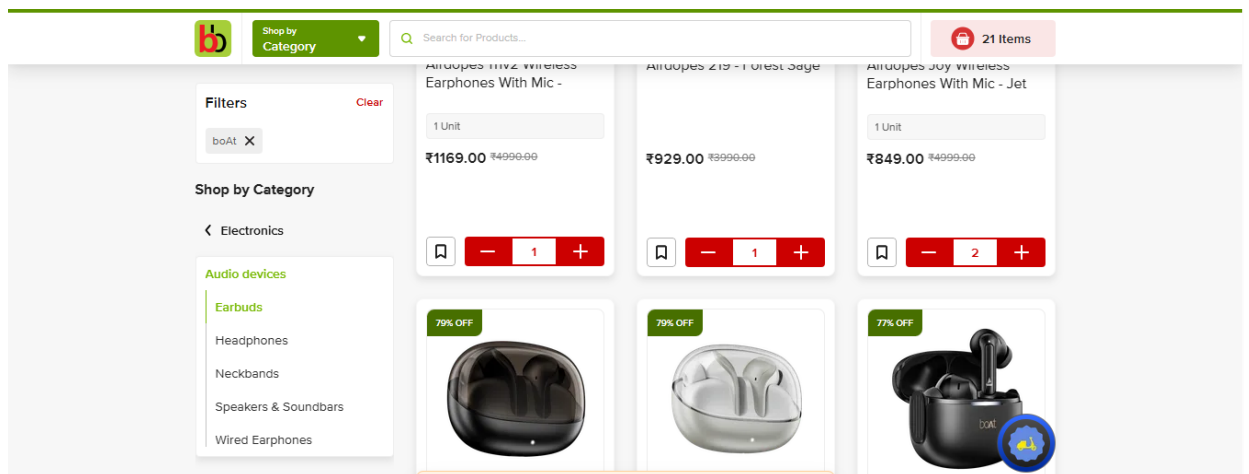
- Earbuds section validation



- Brand filter validation

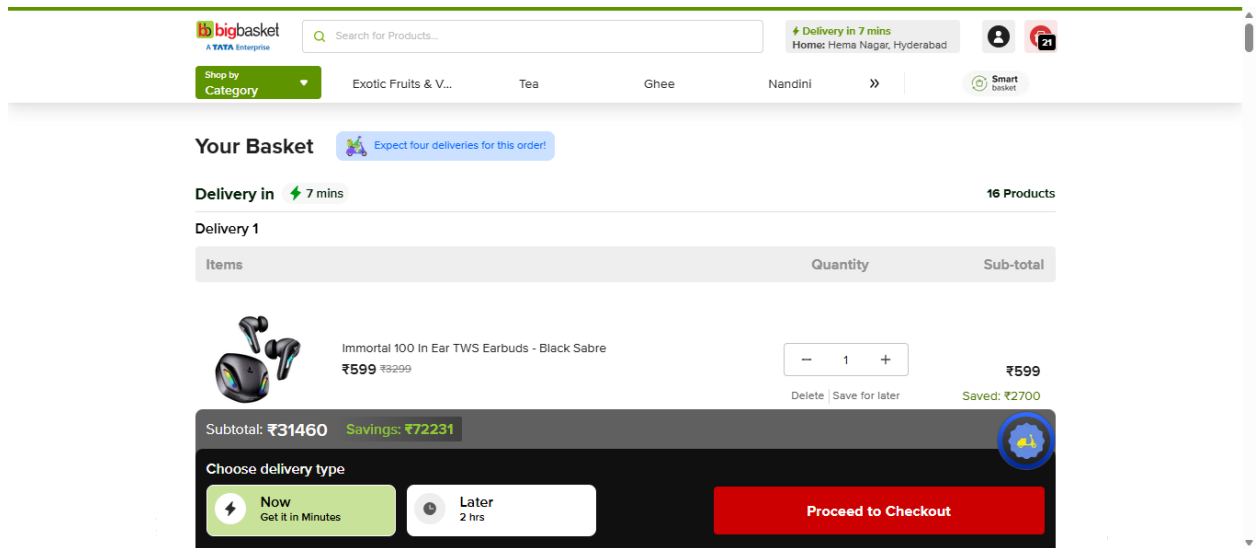


- Add product to basket

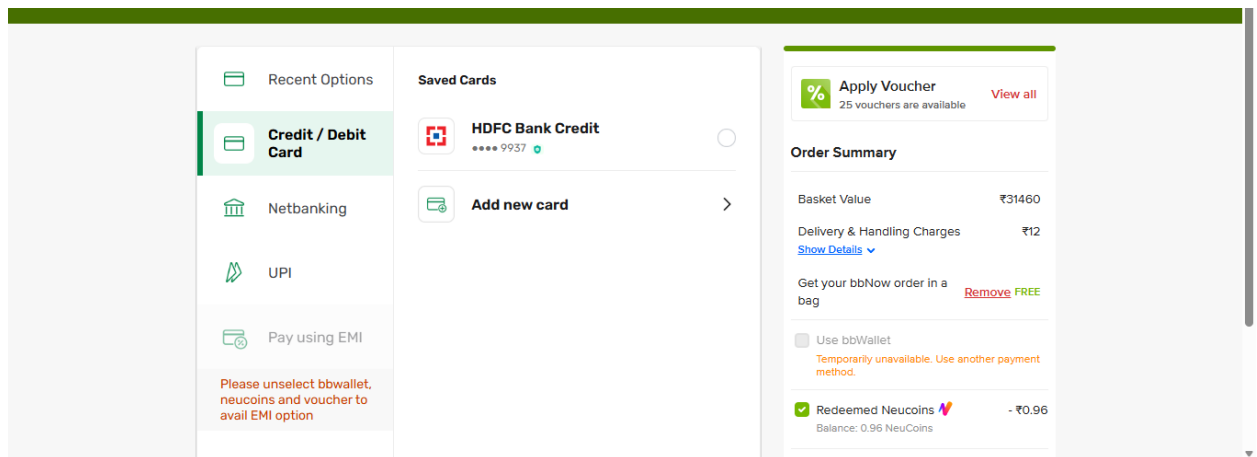


Step 6: Execute Checkout Scenarios

- Navigate to basket



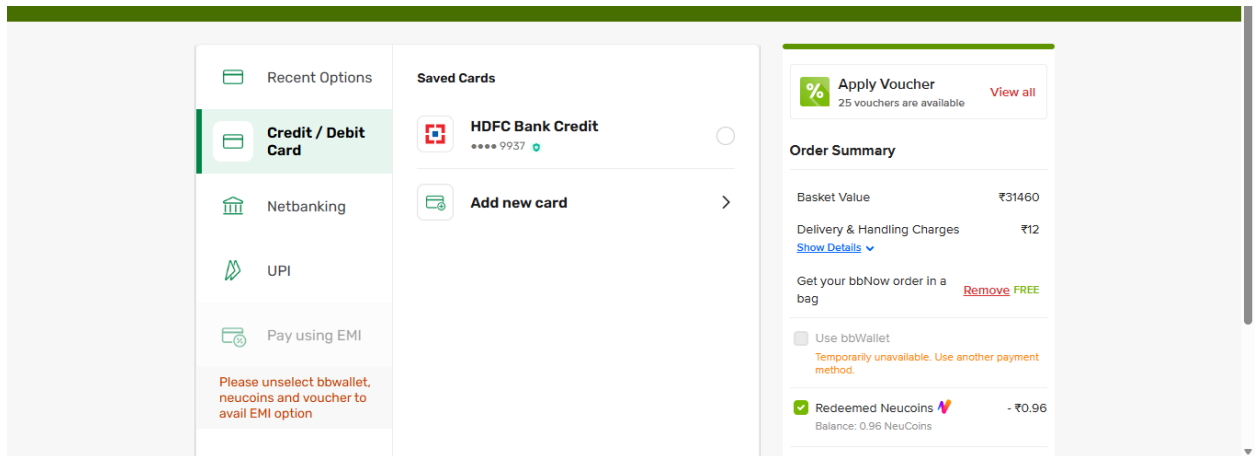
- Proceed to checkout



- Checkout validation

Step 7: Execute End-To-End Flow

- Login
- Electronics navigation
- Product selection
- Basket validation
- Checkout process



Step 8: Assertion Validation

- Assertions validate:
 - URLs
 - Product visibility
 - Error messages
 - Checkout flow

Step 9: Screenshot And Logging

- Screenshots captured for execution steps.
- Logs generated during execution.

LOGS

2026-05-23 17:33:15 - INFO - Launching browser: edge
2026-05-23 17:33:20 - INFO - BigBasket website opened
2026-05-23 17:33:46 - INFO - Login successful
2026-05-23 17:33:57 - INFO - Shop By Category opened
2026-05-23 17:34:07 - INFO - Electronics category opened
2026-05-23 17:34:14 - INFO - Audio devices opened
2026-05-23 17:34:22 - INFO - Earbuds section opened
2026-05-23 17:34:49 - INFO - Boat brand selected
2026-05-23 17:34:57 - INFO - Product added to basket
2026-05-23 17:35:06 - INFO - Basket opened
2026-05-23 17:35:19 - INFO - Checkout page opened
2026-05-23 17:35:23 - INFO - Screenshot attached to Allure report
2026-05-23 17:35:23 - INFO - Automation logs attached to Allure report

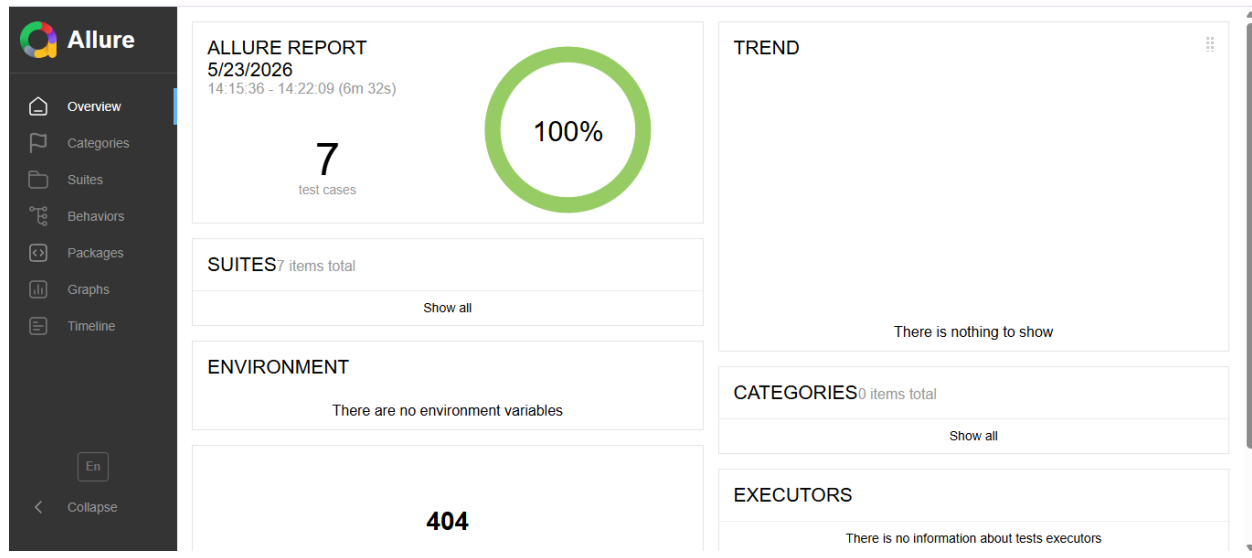
Negative Validation Logs:

2026-05-23 17:35:34 - INFO - Invalid mobile number entered
2026-05-23 17:35:34 - INFO - Verified Continue button is disabled
2026-05-23 17:35:34 - INFO - Invalid mobile test passed
2026-05-23 17:35:54 - INFO - Invalid OTP error message displayed
2026-05-23 17:35:54 - INFO - Invalid OTP test passed

Positive Validation Logs:

2026-05-23 17:36:19 - INFO - Login test passed
2026-05-23 17:37:02 - INFO - Open electronics category test passed
2026-05-23 17:37:58 - INFO - Apply filter test passed
2026-05-23 17:39:05 - INFO - Add product test passed

ALLURE REPORTS



The 'Suites' section displays a table of test results. The table is sortable by order, name, duration, and status. A status bar at the top shows counts for different result types: 0 failed, 0 skipped, 7 passed, 0 pending, and 0 unknown. The table lists 7 test cases, all of which passed successfully.

order	name	duration	status
#7	Add product to basket	1m 06s	passed
#6	Apply boAt brand filter	1m 35s	passed
#1	Complete electronics product checkout flow	2m 03s	passed
#4	Login with valid mobile number	27s 126ms	passed
#5	Open electronics category	40s 145ms	passed
#2	Validate invalid mobile number	12s 119ms	passed
#3	Validate invalid OTP	28s 206ms	passed

Overview

Categories

Suites

Behaviors

Packages

Graphs

Timeline

En

Collapse

Suites

order ↑ name duration ↓ status ↑ Status: 0 0 7 0 0

Marks: [Icons]

✓ #7	Add product to basket	1m 06s
✓ #6	Apply boAt brand filter	1m 35s
✓ #1	Complete electronics product checkout flow	2m 03s
✓ #4	Login with valid mobile number	27s 126ms
✓ #5	Open electronics category	40s 145ms
✓ #2	Validate invalid mobile number	12s 119ms
✓ #3	Validate invalid OTP	28s 206ms

BigBasket End To End Scenario: Complete electronics product checkout fl...

Passed Complete electronics product checkout flow

Overview History Retries

Severity: normal

Duration: 2m 03s

Execution

Test body

When user clicks on login button

And user enters valid mobile number from csv

And user clicks on continue button

And user waits for manual OTP entry

And user clicks on verify button

And user handles got it popup

And user clicks on shop by category

And user clicks on electronics category

And user clicks on audio devices

And user clicks on earbuds category

And user adds first earbud two times

Overview

Categories

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En

Collapse

Suites

order ↑ name duration ↓ status ↑ Status: 0 0 7 0 0

Marks: [Icons]

✓ #7	Add product to basket	1m 06s
✓ #6	Apply boAt brand filter	1m 35s
✓ #1	Complete electronics product checkout flow	2m 03s
✓ #4	Login with valid mobile number	27s 126ms
✓ #5	Open electronics category	40s 145ms
✓ #2	Validate invalid mobile number	12s 119ms
✓ #3	Validate invalid OTP	28s 206ms

And user clicks on shop by category

And user clicks on electronics category

And user clicks on audio devices

And user clicks on earbuds category

And user adds first earbud two times

And user clicks on brands filter

And user selects boAt brand

And user clicks on add button

And user opens basket

And user increments product quantity

And user clicks on checkout button

Then checkout page should be opened

Logs

2026-05-23 14:15:36,865 - INFO -

2026-05-23 14:15:36,866 - INFO - Starting scenario: Complete electronics product checkout flow

2026-05-23 14:15:36,866 - INFO - Browser fetched

2026-05-23 14:15:36,866 - INFO - Base URL fetched

2026-05-23 14:15:36,866 - INFO - Implicit wait fetched

2026-05-23 14:15:36,866 - INFO - Headless mode : False

2026-05-23 14:15:44,571 - INFO - Browser launched successfully

2026-05-23 14:15:45,562 - INFO - Login button clicked successfully

2026-05-23 14:15:45,562 - INFO - Login button clicked

Overview

Categories

Suites

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Collapse

Suites

order ↑ name duration ↓ status ↓

Status: 0 0 7 0 0

Marks: [Icons]

✓ #7	Add product to basket	1m 06s
✓ #6	Apply boAt brand filter	1m 35s
✓ #1	Complete electronics product checkout flow	2m 03s
✓ #4	Login with valid mobile number	27s 126ms
✓ #5	Open electronics category	40s 145ms
✓ #2	Validate invalid mobile number	12s 119ms
✓ #3	Validate invalid OTP	28s 206ms

BigBasket Negative Scenarios: Validate invalid mobile number

Passed Validate invalid mobile number

Overview History Retries

Severity: normal

Duration: 12s 119ms

Execution

Test body

- When user clicks on login button 1 sub-step, 1 attachment 1s 648ms
- And user enters invalid mobile number "12121" 1 sub-step, 1 attachment 450ms
- Then continue button should be disabled 1 sub-step, 2 attachments 463ms

Logs 1.2 KIB

- 2026-05-23 14:17:40,592 - INFO -
- 2026-05-23 14:17:40,592 - INFO - Starting scenario: Validate invalid mobile number
- 2026-05-23 14:17:40,593 - INFO - Browser fetched
- 2026-05-23 14:17:40,593 - INFO - Base URL fetched
- 2026-05-23 14:17:40,593 - INFO - Implicit wait fetched
- 2026-05-23 14:17:40,593 - INFO - Headless mode : False
- 2026-05-23 14:17:47,670 - INFO - Browser launched successfully
- 2026-05-23 14:17:48,816 - INFO - Login button clicked successfully
- 2026-05-23 14:17:48,816 - INFO - Login button clicked
- 2026-05-23 14:17:49,319 - INFO - Screenshot saved at:

Overview

Categories

Suites

Behaviors

Packages

Graphs

Timeline

En

Collapse

Suites

order ↑ name duration ↓ status ↓

Status: 0 0 7 0 0

Marks: [Icons]

✓ #7	Add product to basket	1m 06s
✓ #6	Apply boAt brand filter	1m 35s
✓ #1	Complete electronics product checkout flow	2m 03s
✓ #4	Login with valid mobile number	27s 126ms
✓ #5	Open electronics category	40s 145ms
✓ #2	Validate invalid mobile number	12s 119ms
✓ #3	Validate invalid OTP	28s 206ms

BigBasket Negative Scenarios: Validate invalid OTP

Passed Validate invalid OTP

Overview History Retries

Severity: normal

Duration: 28s 206ms

Execution

Test body

- When user clicks on login button 1 sub-step, 1 attachment 1s 219ms
- And user enters valid mobile number from csv 1 sub-step, 1 attachment 490ms
- And user clicks on continue button 1 sub-step, 1 attachment 386ms
- And user waits for manual invalid OTP entry 1 sub-step, 1 attachment 9s 186ms
- And user clicks on verify button 1 sub-step, 1 attachment 313ms
- Then invalid OTP error message should be displayed 1 sub-step, 3 attachments 8s 445ms

Logs 1.9 KIB

- 2026-05-23 14:17:52,716 - INFO -
- 2026-05-23 14:17:52,717 - INFO - Starting scenario: Validate invalid OTP
- 2026-05-23 14:17:52,717 - INFO - Browser fetched
- 2026-05-23 14:17:52,717 - INFO - Base URL fetched
- 2026-05-23 14:17:52,717 - INFO - Implicit wait fetched

Allure

Overview

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En

Collapse

Suites

order

name

duration

status

Status: 00700

Marks:

✓ #7	Add product to basket	1m 06s	
✓ #6	Apply boAt brand filter	1m 35s	
✓ #1	Complete electronics product checkout flow	2m 03s	
✓ #4	Login with valid mobile number	27s 126ms	
✓ #5	Open electronics category	40s 145ms	
✓ #2	Validate invalid mobile number	12s 119ms	
✓ #3	Validate invalid OTP	28s 206ms	

BigBasket Positive Scenarios: Login with valid mobile number

Passed Login with valid mobile number

OverviewHistoryRetries

Severity: normal

Duration: 27s 126ms

Execution

Test body

When user clicks on login button1 sub-step, 1 attachment1s 213ms

And user enters valid mobile number from csv1 sub-step, 1 attachment508ms

And user clicks on continue button1 sub-step, 1 attachment393ms

And user waits for manual OTP entry1 sub-step, 1 attachment16s 272ms

And user clicks on verify button1 sub-step, 1 attachment334ms

Then user should be logged in successfully1 sub-step, 1 attachment184ms

Logs1.7 KIB

2026-05-23 14:18:20,926 - INFO -
=====

2026-05-23 14:18:20,927 - INFO - Starting scenario: Login with valid mobile number

2026-05-23 14:18:20,927 - INFO - Browser fetched

2026-05-23 14:18:20,927 - INFO - Base URL fetched

2026-05-23 14:18:20,927 - INFO - Implicit wait fetched

2026-05-23 14:18:20,928 - INFO - Headless mode: False

Allure

Overview

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En

Collapse

Suites

order

name

duration

status

Status: 00700

Marks:

✓ #7	Add product to basket	1m 06s	
✓ #6	Apply boAt brand filter	1m 35s	
✓ #1	Complete electronics product checkout flow	2m 03s	
✓ #4	Login with valid mobile number	27s 126ms	
✓ #5	Open electronics category	40s 145ms	
✓ #2	Validate invalid mobile number	12s 119ms	
✓ #3	Validate invalid OTP	28s 206ms	

BigBasket Positive Scenarios: Open electronics category

Passed Open electronics category

OverviewHistoryRetries

Severity: normal

Duration: 40s 145ms

Execution

Test body

When user clicks on login button1 sub-step, 1 attachment1s 643ms

And user enters valid mobile number from csv1 sub-step, 1 attachment577ms

And user clicks on continue button1 sub-step, 1 attachment427ms

And user waits for manual OTP entry1 sub-step, 1 attachment14s 848ms

And user clicks on verify button1 sub-step, 1 attachment320ms

And user handles got it popup1 sub-step, 1 attachment4s 055ms

And user clicks on shop by category1 sub-step, 1 attachment2s 335ms

And user clicks on electronics category1 sub-step, 1 attachment2s 310ms

Then electronics category should be opened1 sub-step, 1 attachment5s 144ms

Logs2.5 KIB

2026-05-23 14:18:48,059 - INFO -
=====

Allure

Overview

Categories

Suites

Behaviors

Packages

Graphs

Timeline

En

Collapse

Suites

order ↑ name duration ↓ status ↓

Status: 0 0 7 0 0

Marks:     

✓ #7 Add product to basket1m 06s

✓ #6 Apply boAt brand filter1m 35s

✓ #1 Complete electronics product checkout flow2m 03s

✓ #4 Login with valid mobile number27s 126ms

✓ #5 Open electronics category40s 145ms

✓ #2 Validate invalid mobile number12s 119ms

✓ #3 Validate invalid OTP28s 206ms

BigBasket Positive Scenarios: Apply boAt brand filter

PassedApply boAt brand filter

OverviewHistoryRetries

Severity: normal

Duration: 1m 35s

Execution

Test body

When user clicks on login button1 sub-step, 1 attachment1s 207ms

And user enters valid mobile number from csv1 sub-step, 1 attachment505ms

And user clicks on continue button1 sub-step, 1 attachment419ms

And user waits for manual OTP entry1 sub-step, 1 attachment56s 326ms

And user clicks on verify button1 sub-step, 1 attachment296ms

And user handles got it popup1 sub-step, 1 attachment5s 105ms

And user clicks on shop by category1 sub-step, 1 attachment2s 318ms

And user clicks on electronics category1 sub-step, 1 attachment2s 324ms

And user opens audio devices from positive flow1 sub-step, 2 attachments2s 451ms

And user opens earbuds category from positive flow1 sub-step, 1 attachment2s 349ms

Allure

Overview

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Packages

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Collapse

Suites

order ↑ name duration ↓ status ↓

Status: 0 0 7 0 0

Marks:     

✓ #7 Add product to basket1m 06s

✓ #6 Apply boAt brand filter1m 35s

✓ #1 Complete electronics product checkout flow2m 03s

✓ #4 Login with valid mobile number27s 126ms

✓ #5 Open electronics category40s 145ms

✓ #2 Validate invalid mobile number12s 119ms

✓ #3 Validate invalid OTP28s 206ms

BigBasket Positive Scenarios: Add product to basket

PassedAdd product to basket

OverviewHistoryRetries

Severity: normal

Duration: 1m 06s

Execution

Test body

When user clicks on login button1 sub-step, 1 attachment1s 594ms

And user enters valid mobile number from csv1 sub-step, 1 attachment632ms

And user clicks on continue button1 sub-step, 1 attachment387ms

And user waits for manual OTP entry1 sub-step, 1 attachment14s 240ms

And user clicks on verify button1 sub-step, 1 attachment335ms

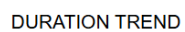
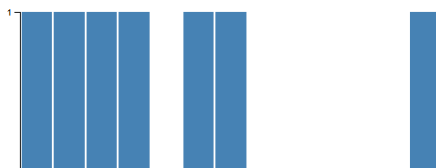
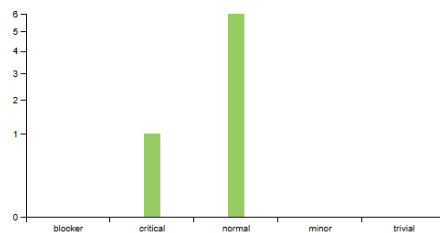
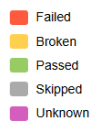
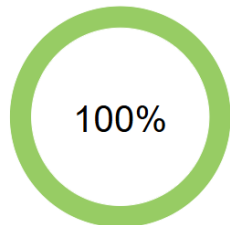
And user handles got it popup1 sub-step, 1 attachment5s 425ms

And user clicks on shop by category1 sub-step, 1 attachment2s 316ms

And user clicks on electronics category1 sub-step, 1 attachment2s 306ms

And user opens audio devices from positive flow1 sub-step, 2 attachments2s 337ms

And user opens earbuds category from positive flow1 sub-step, 1 attachment2s 345ms



TEST REPORT

BDD With Behave

The BDD Automation Testing framework was executed successfully using Behave integrated with Selenium WebDriver. The framework automated both functional and End-To-End scenarios using Gherkin feature files and reusable step definitions.

The framework validated:

- Positive scenarios
- Negative scenarios
- Complete End-To-End shopping flow

using Behavior Driven Development methodology.

Total Scenarios Executed

- Functional Scenarios : 6
 - End-To-End Scenarios : 1
-

Total Test Scenarios

- 7 Scenarios
-

Test Execution Result

- Passed Scenarios : 7
 - Failed Scenarios : 0
 - Skipped Scenarios : 0
-

Accuracy Achieved

- Test Execution Accuracy : 100%

Positive Test Scenarios Executed

- Login Validation
 - Electronics Category Navigation
 - Earbuds Product Validation
 - Apply Brand Filter
 - Add Product To Basket
 - Checkout Navigation
 - End-To-End Shopping Flow
-

Negative Test Scenarios Executed

- Invalid Mobile Number Validation
 - Invalid OTP Validation
-

End-To-End Testing Flow Executed

The End-To-End testing module successfully automated:

- User Login
- Electronics Navigation
- Audio Devices Navigation
- Earbuds Product Selection
- Brand Filter Validation
- Basket Validation
- Checkout Navigation

Reporting And Validation

The framework successfully generated:

- Allure HTML Reports
- Execution Logs
- Step-Level Screenshots
- Assertion Validation Reports

All scenarios executed successfully and the framework validated the complete functionality of the BigBasket web application without failures.

CONCLUSION

The BigBasket Automation Testing Framework was successfully developed and implemented using Selenium WebDriver with Behave BDD framework. The framework automated major functionalities of the BigBasket web application including login validation, Electronics category navigation, Earbuds product filtering, cart operations, and checkout validation.

The project successfully demonstrated:

- Functional Testing
- Negative Testing
- End-To-End Testing
- Assertion Validation
- Reporting And Logging

The framework was designed using the Page Object Model (POM) architecture, which improved code reusability, readability, scalability, and maintainability. Integration of Allure Reports, screenshots, and logging mechanisms provided professional reporting and easier debugging capabilities.

Using Behave framework improved readability of test scenarios through Gherkin syntax and enhanced collaboration between testing and development processes. The End-To-End testing module validated the complete user workflow from login to checkout process successfully.

All automated scenarios executed successfully with high accuracy, validating the stability and functionality of the application. The framework reduced manual testing effort, improved execution efficiency, and provided a scalable automation solution suitable for real-time industry projects.

Overall, the project successfully achieved its objective of developing a reliable, maintainable, and professional BDD automation testing framework for web application testing.